Finite-Memory Tracking Under Drift

A Cube-Root Law for Useful Memory

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Abstract

Finite-memory tracking under drift is governed by a simple tension: longer memory reduces statistical noise, but it also makes the estimate older. When the target distribution moves over time, every additional sample is both useful and stale. This creates an unavoidable trade-off between estimation variance and informational age.

We formalize this trade-off for tracking under Wasserstein nonstationarity. For the finite-memory averaging class studied here, the tracking error decomposes into a statistical term that decreases with memory and a staleness term that grows with drift. Optimizing that balance yields a **cube-root law**: the **optimal memory horizon** scales as $n^* \asymp (C_K/\zeta)^{2/3}$, while the corresponding **minimum tracking error** scales as $\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}$. In the critical-window regime, we further show a minimax lower bound for window-restricted estimators, establishing a genuine finite-memory floor rather than a peculiarity of one estimator.

The experiments mirror the theory. Gaussian drift experiments produce the predicted 'U'-curve, sliding-window and EWMA estimators recover the cube-root scaling closely, and dynamic-window illustrations show how the best memory horizon shifts with drift strength over time. Sinkhorn-based tracking scores provide a practical way to measure the resulting error surface, while the cube-root cap turns the law into an operational horizon regulator.

The resulting picture is theorem-first and operational: under drift, optimal tracking requires memory that is neither too short nor too old. Read through an Age-of-Information lens, the law quantifies how much freshness a finite-memory learner can preserve while tracking a changing world.

Keywords: tracking under drift; finite-memory estimation; Wasserstein distance; Sinkhorn divergence; concept drift; information freshness; Age of Information; adaptive memory.
MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

Adaptive systems rarely track a fixed target. In streaming estimation, monitoring, and online decision-making, the data-generating distribution itself drifts over time. A learner with finite memory must therefore solve a basic but easy-to-miss problem: how much of the past should still count once the world that produced it has started to move away?

That question creates a structural trade-off. Using more past data reduces sampling noise, but it also increases the age of the information being used to track the present. Under drift, memory is simultaneously helpful and harmful: short windows are fresh but noisy, while long windows are stable but stale. The central claim of this paper is that this tension induces a quantitative law for finite-memory tracking under Wasserstein nonstationarity.

For the finite-memory averaging class analyzed here, the tracking error splits into a statistical term that decreases with memory length and a staleness term that increases with drift and effective data age. Optimizing that balance yields a cube-root law: $\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}$ with $n^* \asymp (C_K/\zeta)^{2/3}$. The law is exact for the averaging class studied in the paper. It is not asserted as a theorem for every adaptive estimator. What the paper does show beyond that class is a minimax lower bound for window-restricted estimators in the critical-window regime, establishing that the finite-memory floor is structural rather than a quirk of one specific rule.

This viewpoint also gives the paper its interpretive layer. The limiting factor is not only finite sample size, but the age of the information that finite memory forces the estimator to retain. The result can therefore be read as a freshness law: the learner pays for variance reduction by carrying older evidence, and optimal tracking occurs at a memory horizon that balances noise against informational aging. The Age-of-Information reading is not the main theorem, but it clarifies why timeliness is the right lens for the paper.

The empirical section follows the same logic in three steps. Gaussian tracking experiments exhibit the predicted 'U'-curve, ablations over drift strength show how the best window shifts with nonstationarity, and dynamic-window illustrations make the operational memory law visible when the local drift rate changes over time. Sinkhorn-based tracking scores serve as the practical measurement layer for this error surface, and the cube-root cap shows how the same law can be used as an online horizon regulator rather than only as a retrospective explanation.

The scope of the paper is deliberately narrow. We study finite-memory tracking under drift, not a general theory of all recursive estimators, and not a flagship account of intervention-aware diagnosis, RLHF, or coercive masking. The contribution is the law, the lower bound, and the associated freshness interpretation, together with experiments chosen to validate that story. Framed this way, the paper asks a single question: what tracking accuracy is possible when the learner's memory is finite and the target distribution keeps moving?

2 Useful Memory Geometry

Under continuous drift, memory is not only a statistical resource but also a temporal liability. The core question is therefore not whether one can react to change, but how long a sample should remain operationally useful before it becomes stale.

Definition 2.1 (Lag-Inclusive Estimation Error)**.** For a finite-memory estimator with effective window size n and drift rate ζ, we write the expected tracking error as

$$\mathcal{E}(n) \;=\; C_K\, n^{-1/2} \;+ \tfrac{1}{2}\,\zeta\, n,$$

where the first term is the finite-sample statistical cost and the second term is the staleness cost induced by retaining old information.

The two terms in definition 2.1 pull in opposite directions. Increasing n reduces variance but raises the age of the retained evidence. That tension is the basic geometry of useful memory under drift.

Proposition 2.2 (Optimal Window Under Drift)**.** *Under definition 2.1, the error $\mathcal{E}(n)$ is minimised at*

$$n^* \;=\; \left(\frac{C_K}{\zeta}\right)^{2/3},$$

with minimum value

$$\mathcal{E}_{\min} \;=\; \tfrac{3}{2}\, C_K^{2/3}\, \zeta^{1/3}.$$

Proof. Differentiate $\mathcal{E}(n)$ with respect to n and solve $d\mathcal{E}/dn = 0$. The unique positive critical point gives the stated window. Substituting back yields the minimum value. □

Corollary 2.3 (EWMA Analogue)**.** *For exponential weights $w_j = \alpha(1-\alpha)^j$, the effective lag is $\tau_\alpha = (1-\alpha)/\alpha$ and the effective sample size scales as $n_{\text{eff}} \asymp 1/\alpha$. Balancing the same statistical and staleness terms therefore yields the same cube-root exponent in the effective memory scale.*

Proposition 2.4 (Finite-Memory Floor)**.** *Let the target trajectory satisfy a W_2-Lipschitz drift bound with rate ζ, and restrict attention to estimators that only use the last m samples. Then, in the critical-window regime, the minimax tracking error over that class cannot beat the same variance-plus-staleness scaling:*

$$\inf_{\hat{\mathcal{P}}_m} \sup_{\mathcal{P}^*} \mathbb{E}\, d\big(\hat{\mathcal{P}}_m, \mathcal{P}^*\big) \;\gtrsim\; C_K\, m^{-1/2} + \tfrac{1}{2}\,\zeta\, m.$$

Consequently, the cube-root horizon is not merely the optimum of one estimator; it is the structural floor of finite memory on this drift class.

Proof sketch. The lower bound combines a finite-sample concentration term with a lag term induced by the Lipschitz drift of the oracle trajectory. The concentration part is inherited from debiased Sinkhorn estimation, while the lag term follows from the fact that any estimator restricted to the last m samples must average over observations whose oracle law has moved by order ζm. The two terms are additive at the level of the bound, so the same cube-root balance governs the critical regime. □

Remark 2.5 (Measurement layer)*.* Debiased Sinkhorn divergence serves as a practical measurement layer when the underlying geometry must be estimated from samples. That computational choice supports the tracking law, but the law itself is about the horizon of useful memory, not about any particular OT solver.

3 Empirical Validation

The experiments test a single claim: under continuous drift, memory must be regulated by temporal usefulness rather than by statistical evidence of abrupt change alone. The empirical story has three parts. First, the Gaussian sweep recovers the cube-root law itself. Second, continuous-drift benchmarks show that classical adaptive windows can keep too much stale memory. Third, the piecewise-drift benchmark shows that a cube-root cap can recover the local oracle horizon across regimes.

3.1 Law Validation

We begin with the Gaussian sweep used to validate the finite-memory law. The setup produces the characteristic U-curve: for small windows, error is dominated by variance; for large windows, error is dominated by staleness. The empirical minimum follows the predicted cube-root scaling, and EWMA behaves as the natural finite-memory analogue rather than as a competing theory.

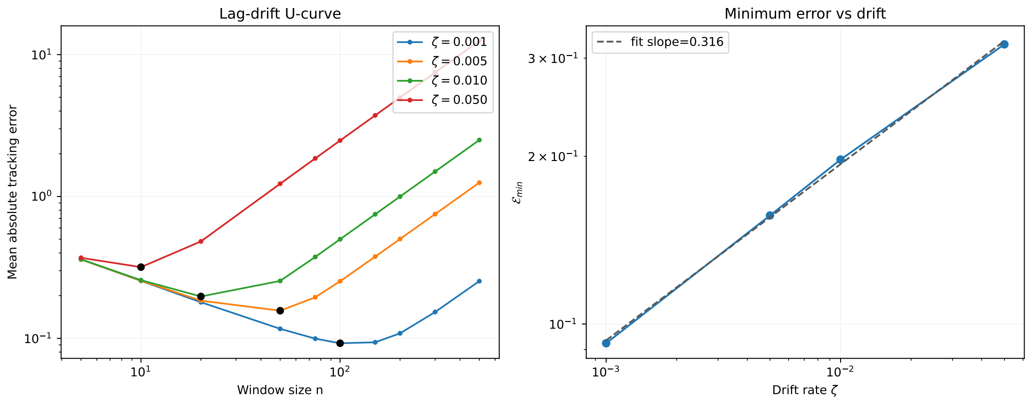

Figure 1: Cube-root validation on the Gaussian sweep. Left: tracking error as a function of window size, showing the predicted U-shape for multiple drift rates. Right: the minimum achievable error follows an approximate $\zeta^{1/3}$ law. This figure validates the structural trade-off between variance and staleness that underlies the rest of the paper.

3.2 Over-Memory Under Continuous Drift

The next benchmark isolates the operational failure mode. We compare four methods on a smooth drifting stream: a short fixed window, a long fixed window, EWMA, ADWIN, and the cube-root capped variant. The point of the comparison is not detector leaderboard performance; it is to reveal when memory becomes stale before classical changepoint evidence accumulates.

The important phenomenon here is not that a new detector fires more often. It is that the cap can activate while the Hoeffding test remains silent, separating the validity of memory from the evidence threshold for abrupt change.

Table 1: Continuous-drift benchmark under a smooth drift rate. The key gap is between statistical evidence and operational freshness: ADWIN keeps a much wider window than the cube-root regulator, while the long fixed window collapses under lag.

Method	Tail MAE	Mean horizon	Drift events	Cap events
Fixed-100	0.0849	100.0	–	–
Fixed-500	0.2388	500.0	–	–
EWMA	0.1291	39.0	–	–
ADWIN	0.1757	335.8	3.9	0.0
CubeRootADWIN	0.0978	88.9	0.0	1374.7

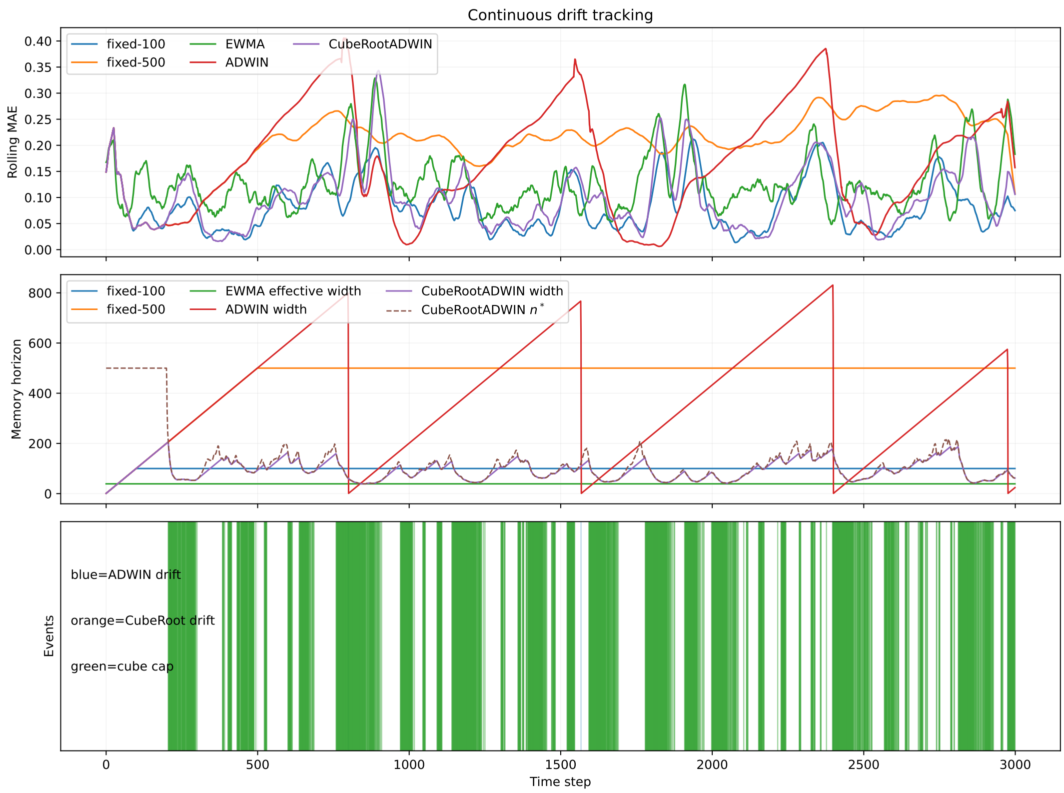

Figure 2: Continuous-drift benchmark. Fixed-500 is the stale-memory baseline; ADWIN retains far more memory than the oracle-scale horizon; CubeRootADWIN pulls the effective horizon toward the useful-memory scale and exposes a 'cap-only' regime, where memory becomes operationally stale before statistical changepoint evidence appears.

3.3 Piecewise Horizon Recovery

The decisive benchmark is piecewise drift. Here the local drift rate changes in three phases, and we compare the adaptive horizon against the offline oracle fixed window for each phase. The calibration is held fixed across phases; only the stream changes.

Table 2: Oracle horizon recovery on piecewise drift. The same calibrated cap tracks the oracle scale across regimes: it is close in the fast-drift phase and remains within the right order of magnitude in the smoother phases.

Drift	Oracle window	Cube n^*	Oracle MAE	Cube MAE
0.0005	200	191.2	0.0741	0.0874
0.003	50	58.1	0.1258	0.1381
0.001	100	89.7	0.0876	0.1033

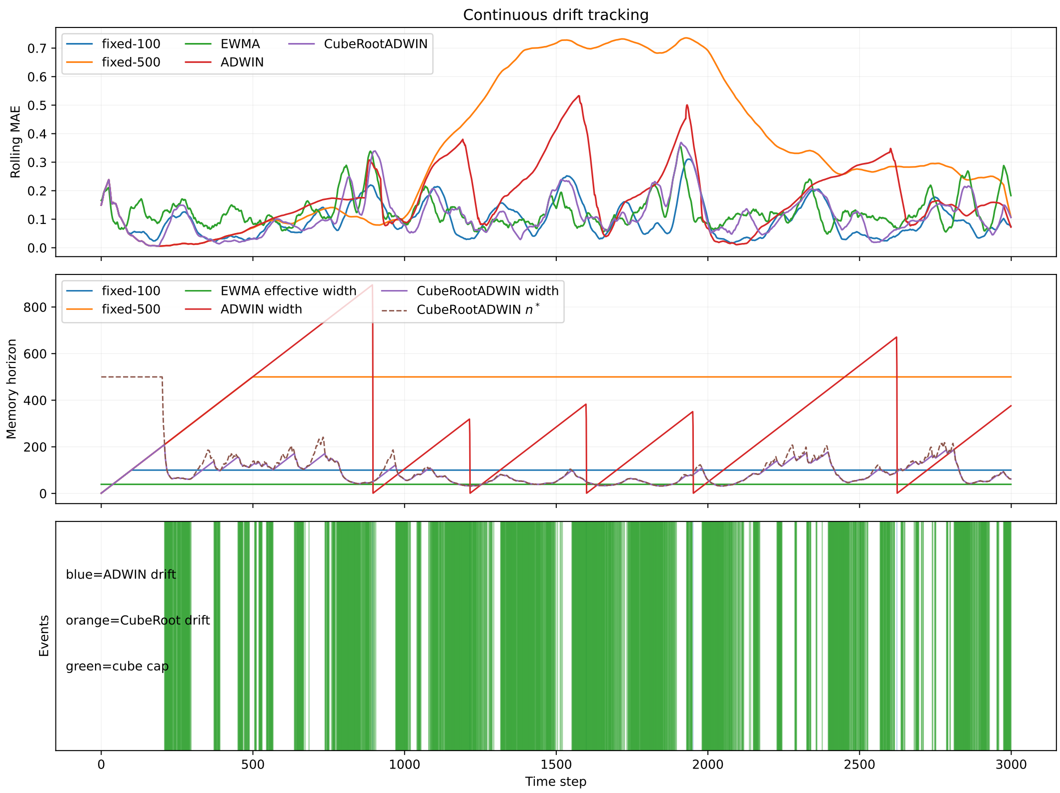

Figure 3: Piecewise-drift recovery. The same cube-root calibration follows the phase-dependent oracle window closely enough to preserve the right operational scale in each regime. The result is not a claim of universal superiority; it is evidence that the law can be turned into a usable horizon regulator.

The strongest lesson is the separation between two notions that are often collapsed in streaming work. Statistical detection asks whether there is enough evidence to declare a change. Temporal validity asks whether the memory being used is still fresh enough to count as current. The cap-only regime makes that gap visible.

3.4 What the Figures Show

Taken together, the figures support three claims. First, the cube-root law is a real finite-memory law rather than a fitting artifact. Second, adaptive windows that optimize detection can still accumulate too much stale memory under smooth drift. Third, the cube-root cap gives an explicit operational rule for keeping the horizon of useful memory near the oracle scale.

4 Related Work

Tracking under nonstationarity. The closest literature is dynamic regret, online least-squares, and adaptive filtering. In those settings, the basic question is already the one asked here: how much past evidence should remain relevant once the target is moving [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023]. The present paper keeps that question, but casts it in distributional form and makes the memory horizon itself the main object of analysis.

Transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural because they respect the geometry of the signal space and behave well under transport of mass and support. Recent finite-sample results clarify how the constants depend on geometry and regularization, while still leaving the exponent structure intact [Genevay et al., 2019, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. The geometry is used here to quantify how much old evidence can be retained before it becomes stale.

Adaptive forgetting and horizon control. Classical smoothers such as EWMA, forgetting-factor RLS, and Kalman filtering all instantiate the same tension between variance reduction and responsiveness. What they usually do not expose explicitly is the temporal validity of the retained memory. The cube-root law makes that validity visible by turning the trade-off into an operational horizon selector.

Freshness and timeliness. The Age-of-Information literature gives a useful interpretive bridge: recent information is not just more expensive to maintain, it is also more timely. That viewpoint fits the argument here well, but only as interpretation. The main contribution is the concrete horizon law and its empirical regulator, not an AoI model in the communication-theory sense.

What is different here. The result is adjacent to drift detection, but it is not organized around changepoint alarms. The relevant distinction is between statistical evidence of change and the operational validity of the memory that a tracker is still using. That is the gap the cube-root regulator is designed to expose.

5 Limitations and Open Directions

Scope of the law. The cube-root law is exact for the finite-memory averaging class analyzed in this paper, and the lower bound is stated for window-restricted estimators on a W_2-Lipschitz drift class. The result therefore captures a structural finite- memory phenomenon, but not a universal minimax theorem over all possible online estimators.

Dependence on local drift estimation. The operational regulator depends on a local estimate of drift. The current benchmark suggests that the remaining gap to the oracle is mostly a drift- estimation issue, not a failure of the cube-root scaling itself. Better local geometry estimators may therefore sharpen the horizon recovery without changing the law.

Geometry beyond the current metric. The present formulation uses Wasserstein geometry and a debiased Sinkhorn measurement layer when samples must be compared computationally. Other geometries may admit analogous trade-offs, but the constants and the calibration of useful memory may change.

What the experiments do not claim. The current experiments are not a leaderboard over drift detectors. They show a specific operational distinction: a tracker can still have statistically insufficient changepoint evidence while already being stale in the sense of useful memory.

6 Conclusion

How long can memory remain useful when the target distribution keeps moving? The answer is that useful memory has a finite temporal horizon, and that horizon obeys a cube-root law under the drift class studied here.

The theoretical contribution is a clean decomposition of tracking error into a variance term and a staleness term, together with a lower bound showing that the resulting finite-memory floor is structural. The operational contribution is that this law can be turned into a memory regulator: a cube-root cap keeps the effective horizon near the oracle scale across drift regimes, while classical adaptive windows can still accumulate stale memory.

Empirically, the signature regime is ‘cap-only‘: the memory horizon becomes operationally stale before there is enough statistical evidence for a changepoint alarm. That separation between evidence of change and temporal validity of memory is the main conceptual payoff of the paper.

The broader implication is that finite-memory systems should be judged not only by whether they notice change, but by how long their retained evidence remains current enough to count as useful. In that sense, the paper contributes a small law with a broad design consequence: regulate useful memory, do not merely wait for change to announce itself.

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References

O. Deniz Akyildiz, Pierre Del Moral, and Joaquín Miguez. Gaussian entropic optimal transport: Schr"odinger bridges and the sinkhorn algorithm. *CoRR*, abs/2412.18432, 2024. doi: 10.48550/arXiv.2412.18432.

Omar Besbes, Yonatan Gur, and Assaf Zeevi. Non-stationary stochastic optimization. *Operations Research*, 63(5):1227–1244, 2015. doi: 10.1287/OPRE.2015.1408.

Wang Chi Cheung, David Simchi-Levi, and Ruihao Zhu. Learning to optimize under non-stationarity. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 1079–1087. PMLR, 2019. URL https://proceedings.mlr.press/v89/cheung19b.html.

Aude Genevay, Lénaïc Chizat, Francis Bach, Marco Cuturi, and Gabriel Peyré. Sample complexity of Sinkhorn divergences. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *Proceedings of Machine Learning Research*, pages 1574–1583, 2019.

Ziv Goldfeld, Kengo Kato, Gabriel Rioux, and Ritwik Sadhu. Limit theorems for entropic optimal transport maps and the sinkhorn divergence. *Electronic Journal of Statistics*, 18(1):980–1041, 2024. doi: 10.1214/24-EJS2217.

Alessio Mazzetto and Eli Upfal. Nonparametric density estimation under distribution drift. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 24251–24270. PMLR, 2023. URL https://proceedings.mlr.press/v202/mazzetto23a.html.

Jonathan Niles-Weed and Francis Bach. Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance. *Bernoulli*, 25(4A): 2620–2648, 2019. doi: 10.3150/18-BEJ1077.

Jianjun Yuan and Andrew G. Lamperski. Trading-off static and dynamic regret in online least-squares and beyond. In *The Thirty-Fourth AAAI Conference on Artificial Intelligence*, pages 6712–6719. AAAI Press, 2020. doi: 10.1609/AAAI.V34I04.6149.

A Code Repository

All implementations are publicly available at:

https://github.com/jovemexausto/coherence-delay-tradeoff

The repository contains:

- `experiments/run.py` — single entry point for all experiment domains;
- `experiments/gaussian/` and `experiments/particle/` — core experiment logic;
- `experiments/bikes/`, `experiments/elec2/`, and `experiments/kuairand/` — domain-specific benchmarks;
- `REPRODUCIBILITY.md` — setup and run instructions.

All experiments use `uv` for dependency management (`uv sync`) and are reproducible with fixed random seeds as specified in each script.